Typesetting with AI Assistance

LaTeX: A Professional Document Preparation System

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Agenda

1. Working with an AI Assistant
2. Prompting for LaTeX
3. Typical AI Failures
4. The Verify Loop
5. Integrity and Privacy

What AI Can and Cannot Do

A chatbot such as Claude, ChatGPT or Gemini can write LaTeX code in seconds. It is a useful **assistant**, but not an author.

Good at

- drafting preambles, tables and figures
- turning a description into maths code
- explaining an error message
- naming commands to look up

Weak at

- it never compiles your file
- it cannot see files you have not shown it
- it may invent packages and references
- it states wrong answers just as confidently

The Rule of This Course

You are the author

The AI drafts. **You** compile, check and correct the result, and you must be able to explain every line of the file you hand in.

Everything you learned so far is what makes this possible:

- reading errors and the `.log` file (**Formatting Text**);
- knowing which commands and environments exist (every lecture);
- knowing what the output **should** look like.

AI may be used only in the two AI workshops and, with a statement, in the final project.

A Weak Prompt and a Good Prompt

Weak: the AI must guess everything.

```
make a latex table of the first five primes and their squares
```

Good: engine, what to return, layout and limits.

```
I compile a LaTeX article with pdflatex. Write only the code to paste
inside the document body (no \documentclass, no preamble).
I need a centred table, without a caption, made with the center
environment and tabular. It has two columns with the math-mode headers
$p$ and $p^2$, a vertical rule between the columns, and \hline rules
at the top, below the header and at the bottom. Rows: the first five
prime numbers and their squares. Use no extra packages.
```

What a Good Prompt Says

1. **Engine and class**: pdflatex or xelatex, `article`, `report`, `beamer`, …
2. **What to return**: a snippet for the body, the preamble only, or a whole document.
3. **Allowed packages**: “only amsmath and graphicx”, “no extra packages”.
4. **Exact content**: captions, labels, numbers and text, copied in full.
5. **Layout**: rules, alignment, numbered or unnumbered, where the caption goes.
6. **One piece at a time**: one table or one proof per prompt, then compile.

The more the AI must guess, the more you must check.

Asking About an Error

Give the AI what it cannot see: the engine, the **complete** message and the code.

```
I compile this LaTeX file with pdflatex (via latexmk).
This is the complete error message from the terminal:

[paste every line from the first "!" (or "Runaway argument?")
 down to the ? prompt, including the l.<n> line]

This is the whole file:

[paste the .tex file]

Explain the cause in two sentences.
Then show only the line or lines I need to change. Do not rewrite the
↪  whole file.
```

Asking for **only the changed lines** stops the AI from silently changing other parts of your file.

Packages That Do Not Exist

The AI writes package names that **look** right:

```
\usepackage{amsthm, amsproof}
```

```
! LaTeX Error: File `amsproof.sty' not found.
```

- Check a package before you trust it: `texdoc <name>` in the terminal, or search `https://ctan.org`.
- The same happens with **options** and **commands**: if the compiler does not know it, it does not exist.

Old Habits

AI learned from decades of old LaTeX files, so it often uses commands that still compile but should no longer be used.

AI writes	Use instead	Why
`$$ ... $$`	`\[ ... \]`	plain TeX, wrong spacing
`{\bf text}`	`\textbf{text}`	from LaTeX 2.09 (1980s)
`eqnarray`	`align`	uneven space around $=$
`[H]`	`[ht]`	needs the `float` package
`inputenc` with `utf8`	(nothing)	UTF-8 is the default since 2018

Missing Setup

The AI shows the part you asked for and forgets what it needs.

```
\begin{theorem}[Euclid]
 ...
\end{theorem}
% AI: "amsthm provides theorem."  (wrong)
```

```
! LaTeX Error: Environment theorem undefined.
```

`amsthm` only provides `\newtheorem`. You declare the environment yourself:

```
\newtheorem{theorem}{Theorem}
```

Labels, Engines and Invented Sources

- **Labels**: the AI renames them (`tab:triple` instead of `tab:triples`), so the text shows **??**. latexmk reports

```
Latexmk: Summary of warnings from last run of *latex:
  Latex failed to resolve 1 reference(s)
```

- **Thai**: the AI often suggests `pdflatex` with `inputenc` or `babel`. In this course Thai documents use `thaispec` with `xelatex`.
- **References**: the AI can invent books, articles and DOIs that do not exist. Never copy a `.bib` entry from a chatbot without finding the source yourself.

It Compiles, but It Is Wrong

The most dangerous answers produce **no** error at all.

Wrong maths

```
a^2 + b^2
  &= (m^2 - n^2)^2 + (2mn)^2 \\
  &= m^4 - 2m^2n^2 + n^4
     + 2m^2n^2
```

$(2mn)^2 = 4m^2n^2$, not $2m^2n^2$. The compiler does not check mathematics.

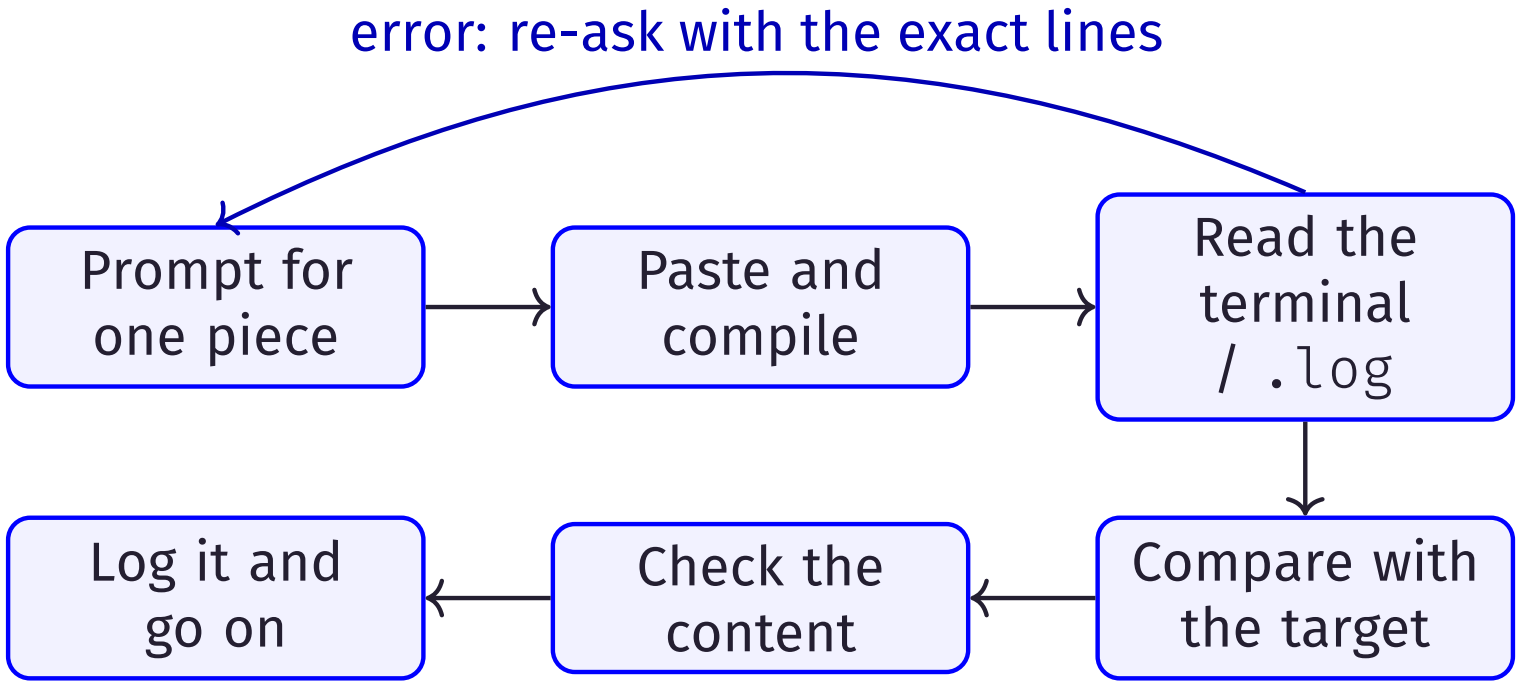
A fix that only hides the error

```
a short script called
$birthday_table.py$
```

The `Missing $ inserted` error is gone, but the output is an italic **birthdaytable.py** with a subscript. The real fix is `birthday\_table.py`.

"It compiles" is where checking starts, not where it ends.

The Verify Loop



- Fix errors **one at a time**, from the first one in the terminal.
- When you re-ask, paste the **new** message; do not just say "it still fails".

Before You Accept an Answer

1. Does it compile with no errors and no **new** warnings?
2. Did the AI do what the prompt asked (packages, labels, layout)?
3. Does the output match the target, side by side?
4. Is the content right: every number, every step of the maths, every source?
5. Can you explain every line? If not, ask the AI to explain, then check that too.

Verification log

Write down what the AI got wrong and how you found it. In the AI workshops this log is part of what you hand in.

Using AI in Your Project

AI is allowed in the final project if you say how you used it. Add a short statement at the end of the document:

```
\section*{AI Use Statement}
I used a chatbot to draft the table in Section 2
and to explain two compile errors. I checked every
number in the table against my data and rewrote
the proof in Section 3 myself.
```

- Name the tool, what you used it for and what you checked.
- You are responsible for every line, including those the AI wrote.
- If you cannot explain a line, rewrite it yourself or remove it.

Privacy

Whatever you paste into a chatbot leaves your computer and may be stored.

Never paste

- your student ID, name, or anyone else's personal data;
- grades, marks or other people's work;
- unpublished data or text you are not allowed to share.

Work in a file with a generic name such as `main.tex`, and rename it with your student ID only when you hand it in.

Summary

- Tell the AI the engine, what to return, the packages and the exact content.
- Ask for one piece at a time, and only the lines that change.
- Expect invented packages, old habits, missing setup, renamed labels and wrong maths.
- Verify: compile, read the log, compare with the target, check the content.
- Keep a log, disclose AI use in the project, and never paste personal data.

Workshops: **Drafting with AI** and **Debugging with AI**.

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